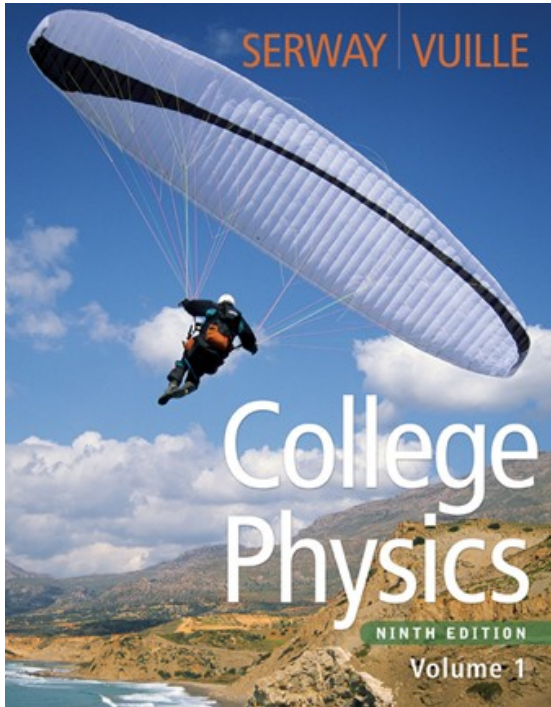




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Chris Vuille

Chapter Six

Momentum and Collisions

Collisions

- Conservation of momentum allows complex collision problems to be solved without knowing about the forces involved
- Information about the average force can be derived

Momentum

- The linear momentum $\vec{p}$ of an object of mass m moving with a velocity $\vec{v}$ is defined as the product of the mass and the velocity

$$\vec{p} = m\vec{v}$$

–SI Units are kg m / s

–Vector quantity, the direction of the momentum is the same as the velocity's

More About Momentum

- Momentum components

$$p_x = m v_x \text{ and } p_y = m v_y$$

- Applies to two-dimensional motion

- Momentum is related to kinetic energy

$$KE = \frac{p^2}{2m}$$

Impulse

- In order to *change* the momentum of an object, a force must be applied
- The time rate of change of momentum of an object is equal to the net force acting on it

$$\frac{\Delta \vec{p}}{\Delta t} = \frac{m(v_f - v_i)}{\Delta t} = \vec{F}_{net}$$

- Gives an alternative statement of Newton's second law
- Also valid when the forces are not constant

Impulse cont.

- When a single, constant force acts on the object, there is an **impulse** delivered to the object

$$\vec{\mathbf{I}} = \vec{\mathbf{F}}\Delta t$$

$\vec{\mathbf{I}}$ is defined as the *impulse*

Vector quantity, the direction is the same as the direction of the force

SI unit of impulse: $\text{kg} \cdot \text{m} / \text{s}$

Impulse-Momentum Theorem

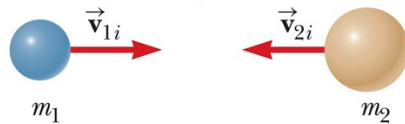
- The theorem states that the impulse acting on the object is equal to the change in momentum of the object

$$\vec{I} = \vec{F}\Delta t = \Delta\vec{p} = m\vec{v}_f - m\vec{v}_i$$

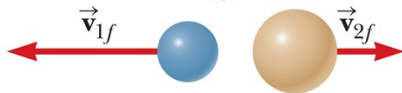
—If the force is not constant, use the *average force* applied

Momentum and Impulse (5 of 9)

Before the collision, these particles have equal and opposite velocities.



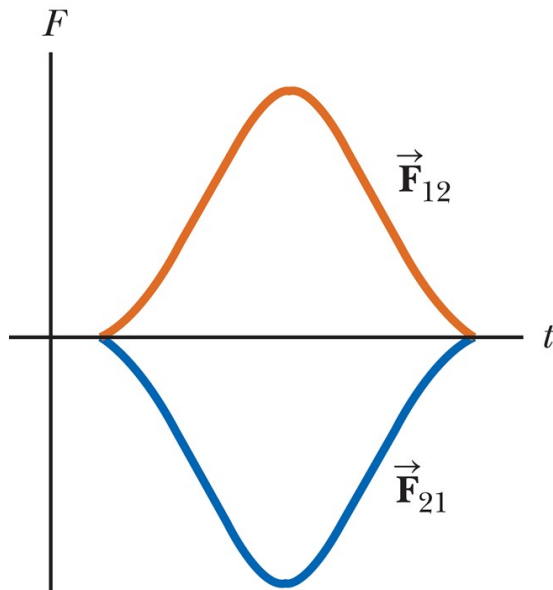
After the collision both velocities change, but the total momentum of the system remains the same.



$$\mathbf{F}_{21}\Delta t = m_1\mathbf{v}_{1f} - m_1\mathbf{v}_{1i}$$

$$\mathbf{F}_{12}\Delta t = m_2\mathbf{v}_{2f} - m_2\mathbf{v}_{2i}$$

Momentum and Impulse (6 of 9)



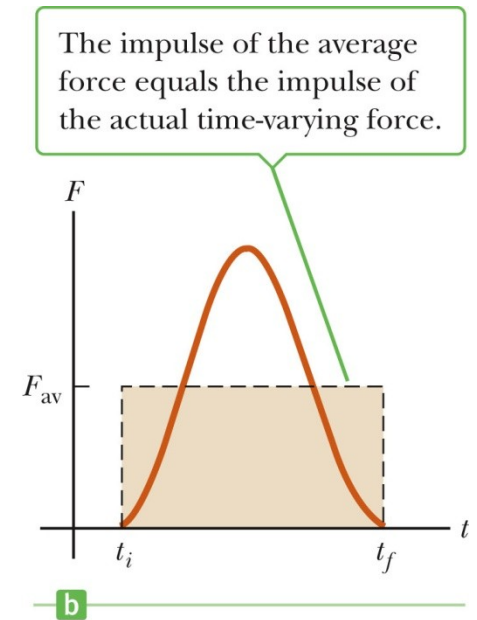
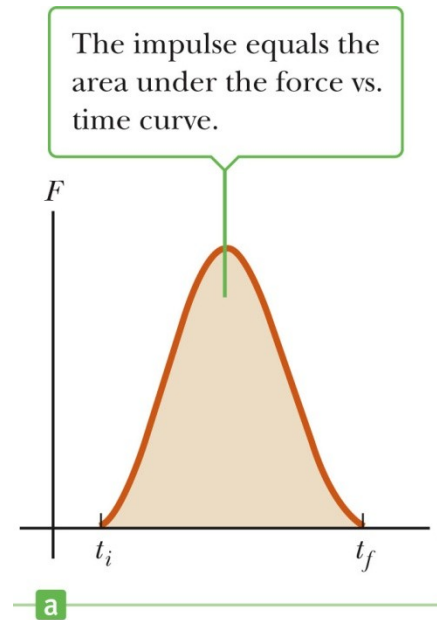
$$\vec{F}_{21}\Delta t = -\vec{F}_{12}\Delta t$$

$$m_1\mathbf{v}_{1f} - m_1\mathbf{v}_{1i} = -(m_2\mathbf{v}_{2f} - m_2\mathbf{v}_{2i})$$

$$m_1\mathbf{v}_{1i} + m_2\mathbf{v}_{2i} = m_1\mathbf{v}_{1f} + m_2\mathbf{v}_{2f}$$

Average Force in Impulse

- The average force can be thought of as the constant force that would give the same impulse to the object in the time interval as the actual time-varying force gives in the interval



Average Force cont.

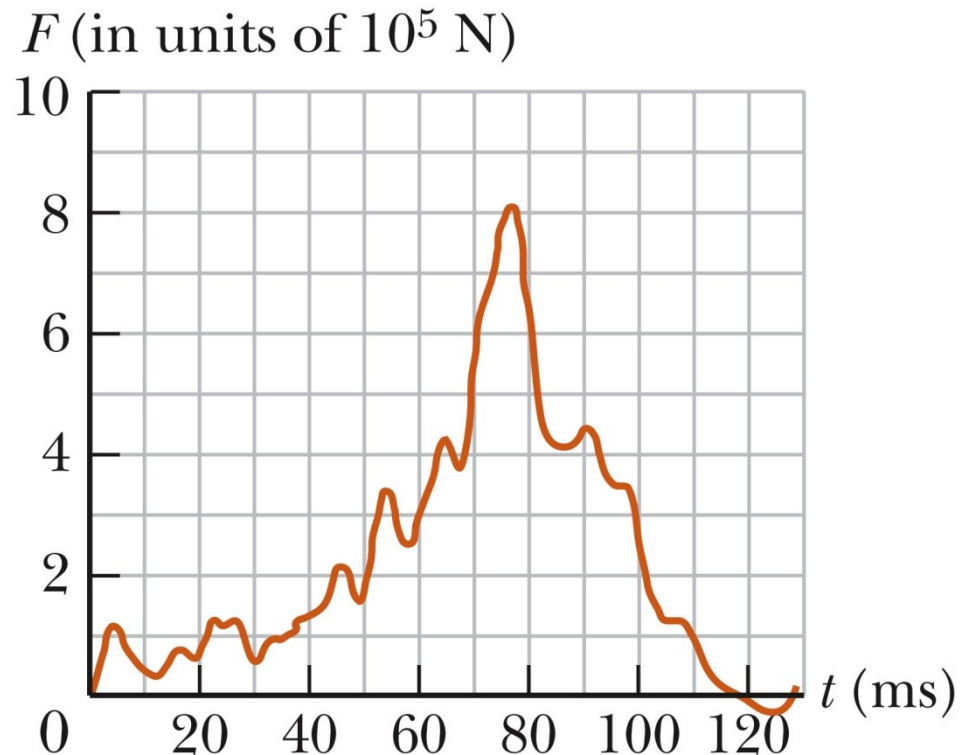
- The impulse imparted by a force during the time interval Δt is equal to the area under the force-time graph from the beginning to the end of the time interval
- Or, the impulse is equal to the average force multiplied by the time interval, $\vec{F}_{av} \Delta t = \Delta \vec{p}$

Impulse Applied to Auto Collisions

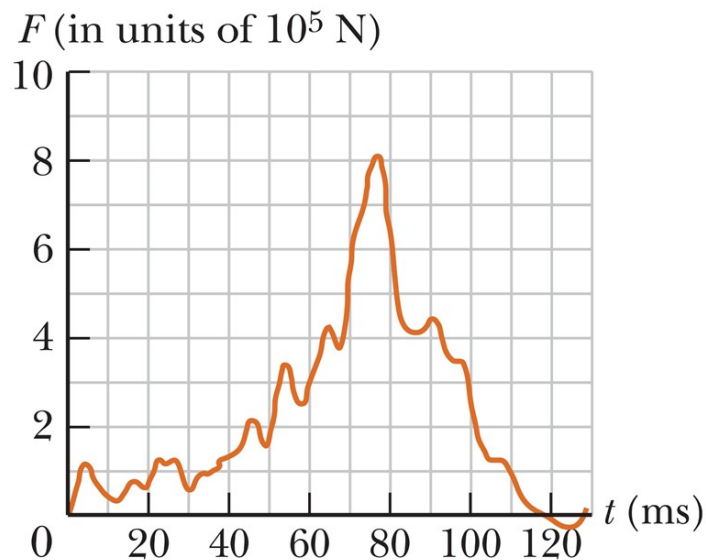
- The most important factor is the collision time, or the time it takes the person to come to a rest
 - Increasing this time will reduce the chance of dying in a car crash
- Ways to increase the time
 - Seat belts
 - Air bags

Typical Collision Values

- For a 75 kg person traveling at 27 m/s (60.0 mph) and coming to stop in 0.010 s
- $F = -2.0 \times 10^5 \text{ N}$
- $a = 280 \text{ g}$
- Almost certainly fatal



Injury in Automobile Collisions (3 of 3)



$$F_{\text{av}} = (75 \text{ kg}) \left(\frac{27 \text{ m/s}}{0.15 \text{ s}} \right) = 1.35 \times 10^4 \text{ N}$$

$$a_{\text{av}} = \left(\frac{27 \text{ m/s}}{0.15 \text{ s}} \right) \div 9.8 \text{ m/s}^2 \approx 18g$$

$$\frac{F_{\text{av}}}{A} = \frac{1.35 \times 10^4 \text{ N}}{0.5 \text{ m}^2} = 2.7 \times 10^4 \text{ N/m}^2$$

Injury in Automobile Collisions (1 of 3)



TRL Ltd./Science Source

b

90 kN: bone fracture

150g, 4 ms, or 50g, 60 ms → fatal 50% of the time

Whole-body impact pressure $\geq 3.4 \times 10^5 \text{ N/m}^2$

→ fatal 50% of the time

Injury in Automobile Collisions (2 of 3)

$$F_{\text{av}} \Delta t = mv_f - mv_i$$

$$F_{\text{av}} = \frac{mv_f - mv_i}{\Delta t} = \frac{0 - (75 \text{ kg})(27 \text{ m/s})}{0.010 \text{ s}} = -2.0 \times 10^5 \text{ N}$$

$$a = \left| \frac{\Delta v}{\Delta t} \right| = \frac{27 \text{ m/s}}{0.010 \text{ s}} = 2700 \text{ m/s}^2 = \frac{2700 \text{ m/s}^2}{9.8 \text{ m/s}^2} g = 280g$$

$$\frac{F_{\text{av}}}{A} = \frac{2.0 \times 10^5 \text{ N}}{0.5 \text{ m}^2} = 4 \times 10^5 \text{ N/m}^2$$

Seat Belts

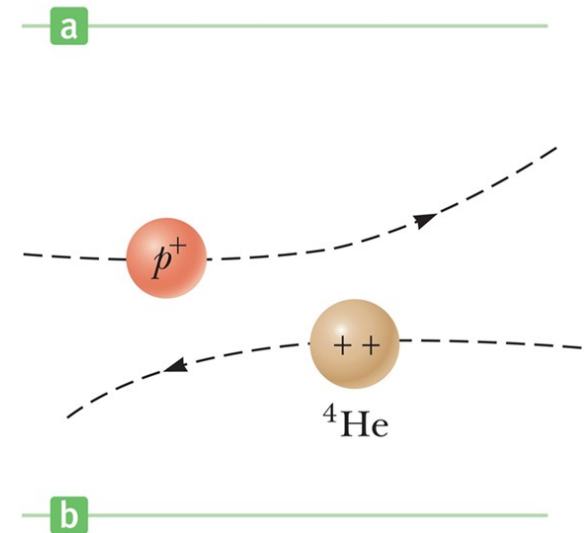
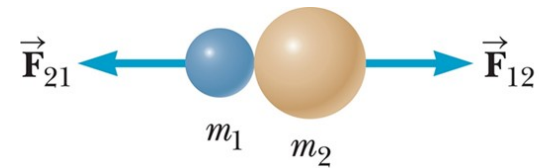
- Seat belts
 - Restrain people so it takes more time for them to stop
 - New time is about 0.15 seconds
 - New force is about 9.8 kN
 - About one order of magnitude below the values for an unprotected collision

Air Bags

- The air bag increases the time of the collision
- It will also absorb some of the energy from the body
- It will spread out the area of contact
 - Decreases the pressure
 - Helps prevent penetration wounds

Conservation of Momentum

- Momentum in an isolated system in which a collision occurs is conserved
 - A collision may be the result of physical contact between two objects
 - “Contact” may also arise from the electrostatic interactions of the electrons in the surface atoms of the bodies
 - An isolated system will have not external forces



Conservation of Momentum, cont

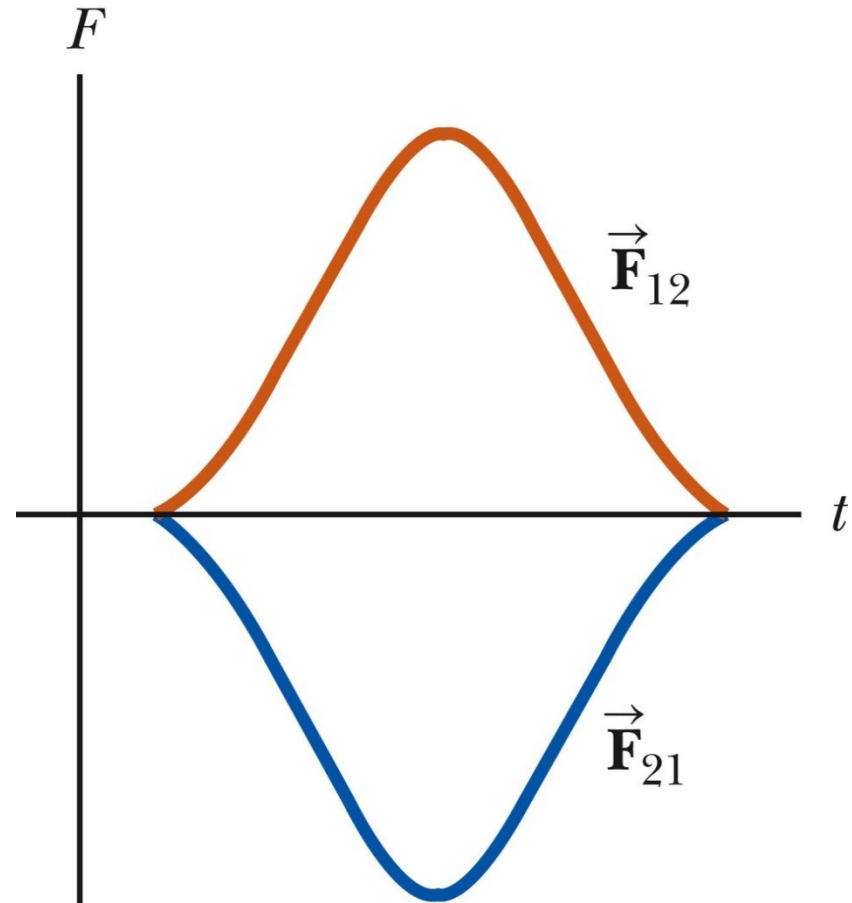
- The principle of conservation of momentum states when no external forces act on a system consisting of two objects that collide with each other, the total momentum of the system remains constant in time
 - Specifically, the total momentum before the collision will equal the total momentum after the collision

Conservation of Momentum, Example

- The momentum of each object will change
- The total momentum of the system remains constant

Forces in a Collision

- The force with which object 1 acts on object 2 is equal and opposite to the force with which object 2 acts on object 1
- Impulses are also equal and opposite



Conservation of Momentum, cont.

- Mathematically:

$$m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i} = m_1 \vec{v}_{1f} + m_2 \vec{v}_{2f}$$

- Momentum is conserved for the *system* of objects
- The system includes all the objects interacting with each other
- Assumes only internal forces are acting during the collision
- Can be generalized to any number of objects

Types of Collisions

- Momentum is conserved in any collision
- Inelastic collisions
 - Kinetic energy is not conserved
 - Some of the kinetic energy is converted into other types of energy such as heat, sound, work to permanently deform an object
 - Perfectly inelastic collisions occur when the objects stick together
 - Not all of the KE is necessarily lost

More Types of Collisions

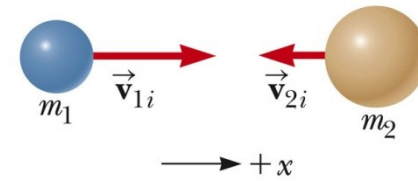
- Elastic collision
 - Both momentum and kinetic energy are conserved
- Actual collisions
 - Most collisions fall between elastic and perfectly inelastic collisions

Perfectly Inelastic Collisions

- When two objects stick together after the collision, they have undergone a perfectly inelastic collision
- Conservation of momentum becomes

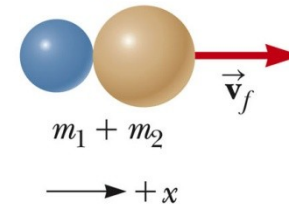
$$m_1 v_{1i} + m_2 v_{2i} = (m_1 + m_2) v_f$$

Before a perfectly inelastic collision the objects move independently.



a

After the collision the objects remain in contact. System momentum *is* conserved, but system energy is *not* conserved.



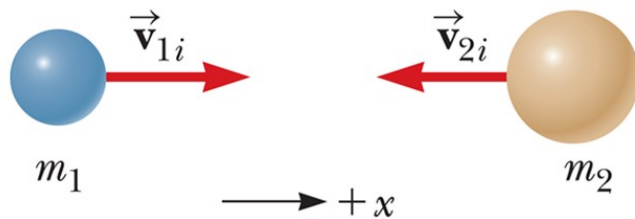
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Some General Notes About Collisions

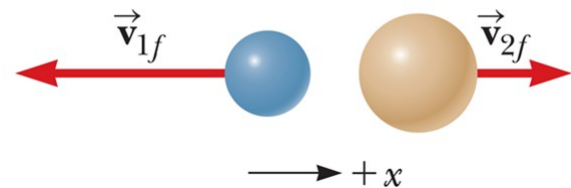
- Momentum is a vector quantity
 - Direction is important
 - Be sure to have the correct signs

More About Elastic Collisions

Before an elastic collision the two objects move independently.



After the collision the object velocities change, but **both** the energy and momentum of the system are conserved.



- Both momentum and kinetic energy are conserved
- Typically have two unknowns (the two final velocities)

$$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$$

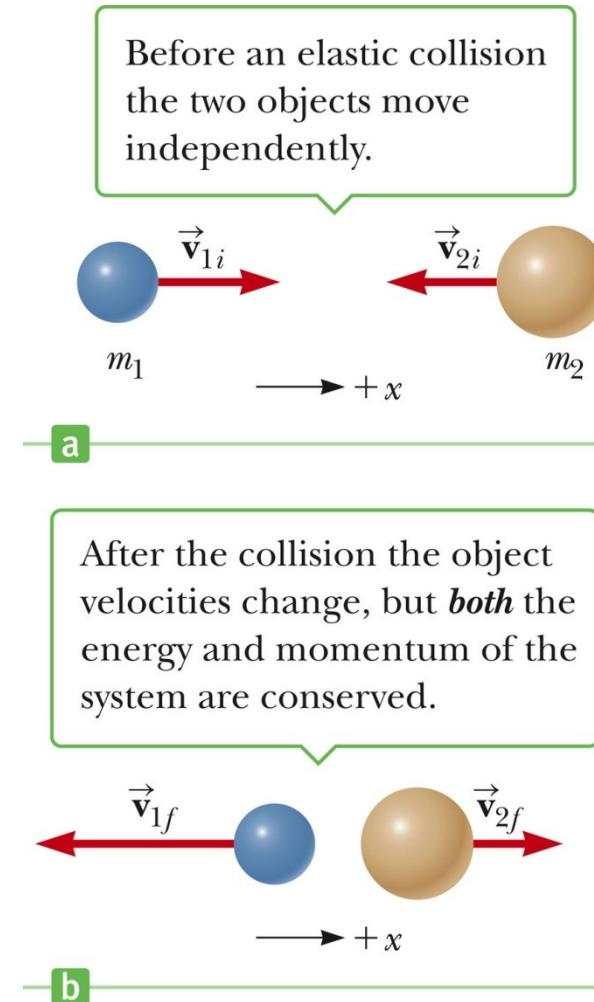
$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$

- Solve the equations simultaneously

Elastic Collisions, cont.

- A simpler equation can be used in place of the KE equation

$$v_{1i} - v_{2i} = -(v_{1f} - v_{2f})$$



Summary of Types of Collisions

- In an elastic collision, both momentum and kinetic energy are conserved
- In an inelastic collision, momentum is conserved but kinetic energy is not
- In a *perfectly* inelastic collision, momentum is conserved, kinetic energy is not, and the two objects stick together after the collision, so their final velocities are the same

Problem Solving for One -Dimensional Collisions

- **Coordinates:** Set up a coordinate axis and define the velocities with respect to this axis
 - It is convenient to make your axis coincide with one of the initial velocities
- **Diagram:** In your sketch, draw all the velocity vectors and label the velocities and the masses

Problem Solving for One -Dimensional Collisions, 2

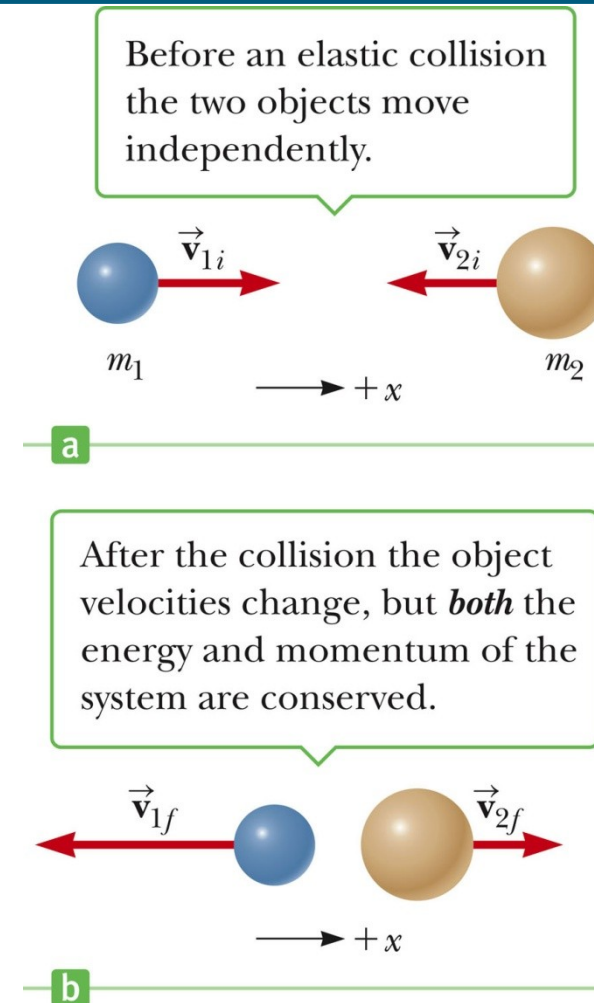
- **Conservation of Momentum:** Write a general expression for the total momentum of the system *before* and *after* the collision
 - Equate the two total momentum expressions
 - Fill in the known values

Problem Solving for One -Dimensional Collisions, 3

- **Conservation of Energy:** If the collision is elastic, write a second equation for conservation of KE, or the alternative equation
 - This only applies to perfectly elastic collisions
- **Solve:** the resulting equations simultaneously

Sketches for Collision Problems

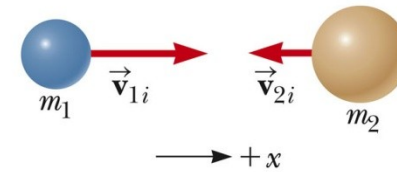
- Draw “before” and “after” sketches
- Label each object
 - Include the direction of velocity
 - Keep track of subscripts



Sketches for Perfectly Inelastic Collisions

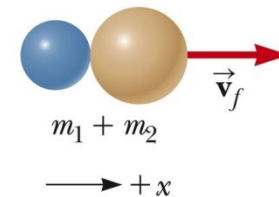
- The objects stick together
- Include all the velocity directions
- The “after” collision combines the masses
- Both move with the same velocity

Before a perfectly inelastic collision the objects move independently.



a

After the collision the objects remain in contact. System momentum *is* conserved, but system energy is *not* conserved.



b

Glancing Collisions

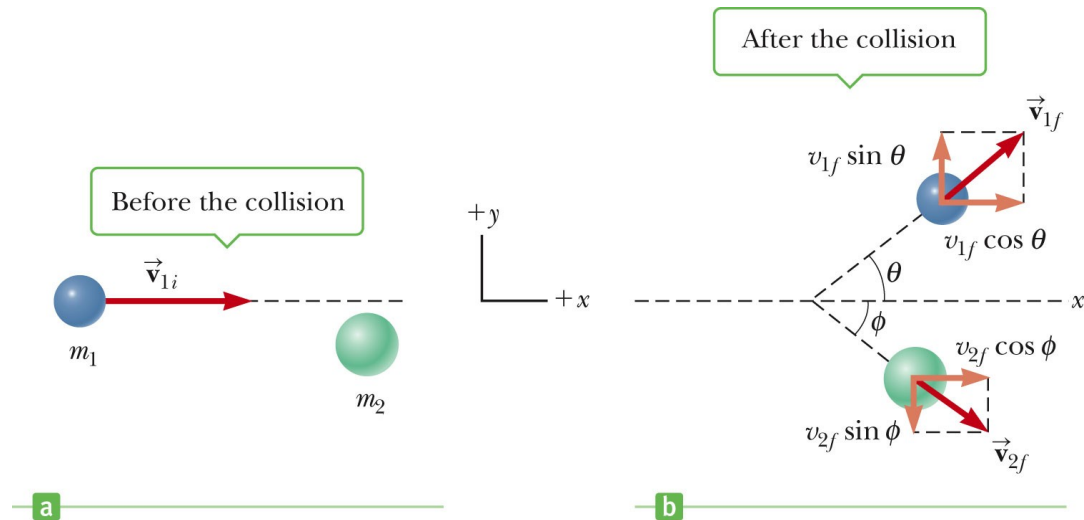
- For a general collision of two objects in three-dimensional space, the conservation of momentum principle implies that the *total momentum of the system in each direction is conserved*

$$m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx} \text{ and}$$

$$m_1 v_{1iy} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$$

- Use subscripts for identifying the object, initial and final velocities, and components

Glancing Collisions – Example



- The “after” velocities have x and y components
- Momentum is conserved in the x direction and in the y direction
- Apply conservation of momentum separately to each direction

Problem Solving for Two-Dimensional Collisions

- **Coordinates:** Set up both coordinate axes and define your velocities with respect to these axes
 - It is convenient to choose the x- or y- axis to coincide with one of the initial velocities
- **Diagram:** In your sketch, draw and label all the velocities and masses

Problem Solving for Two-Dimensional Collisions, 2

- **Conservation of Momentum:** Write expressions for the x and y components of the momentum of each object before and after the collision
 - Write expressions for the total momentum before and after the collision in the x-direction and in the y-direction

Problem Solving for Two-Dimensional Collisions, 3

- **Conservation of Energy:** If the collision is perfectly elastic, write an expression for the total energy before and after the collision
 - Equate the two expressions
 - Fill in the known values
 - Solve the quadratic equations
 - Can't be simplified
 - Remember to skip this step if the collision is not perfectly elastic

Problem Solving for Two-Dimensional Collisions, 4

- **Solve** for the unknown quantities
 - Solve the equations simultaneously
 - There will be two equations for inelastic collisions
 - There will be three equations for elastic collisions

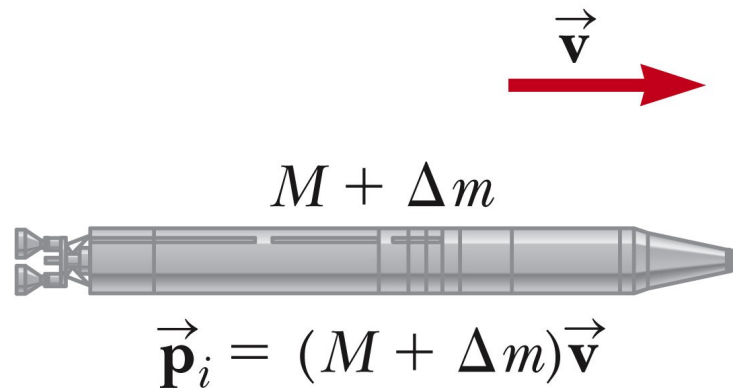
Rocket Propulsion

- The operation of a rocket depends on the law of conservation of momentum as applied to a system, where the system is the rocket plus its ejected fuel
 - This is different than propulsion on the earth where two objects exert forces on each other
 - Road on car
 - Train on track

Rocket Propulsion, 2

- The rocket is accelerated as a result of the thrust of the exhaust gases
- This represents the inverse of an inelastic collision
 - Momentum is conserved
 - Kinetic Energy is increased (at the expense of the stored energy of the rocket fuel)

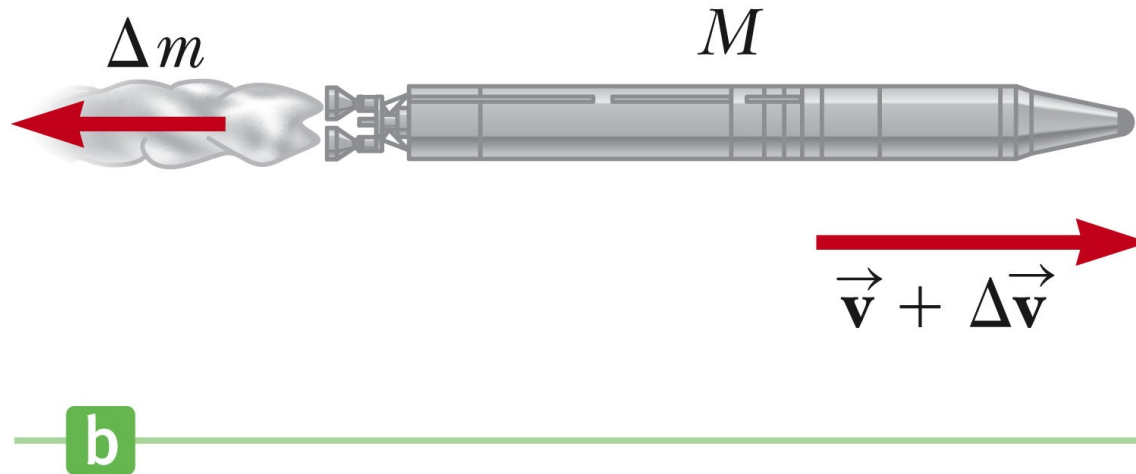
Rocket Propulsion, 3



a

- The initial mass of the rocket is $M + \Delta m$
 - M is the mass of the rocket
 - Δm is the mass of the fuel about to be burned
- The initial velocity of the rocket is $\vec{v}$

Rocket Propulsion



- The rocket's mass is M
- The mass of the fuel, Δm , has been ejected
- The rocket's speed has increased to $\vec{v} + \Delta \vec{v}$

Rocket Propulsion, final

- The basic equation for rocket propulsion is:

$$v_f - v_i = v_e \ln \left(\frac{M_i}{M_f} \right)$$

- M_i is the initial mass of the rocket plus fuel
- M_f is the final mass of the rocket plus any remaining fuel
- The speed of the rocket is proportional to the exhaust speed
- For best results, the exhaust speed should be as high as possible
- Typical exhaust speeds are several kilometers per second

Thrust of a Rocket

- The thrust is the force exerted on the rocket by the ejected exhaust gases
- The instantaneous thrust is given by

$$Ma = M \frac{\Delta v}{\Delta t} = \left| v_e \frac{\Delta M}{\Delta t} \right|$$

- The thrust increases as the exhaust speed increases and as the burn rate ($\Delta M/\Delta t$) increases